

Challenge Description

The challenge presents a message from a musician known as **"Mr-Worldwide."** The attached file contains several pairs of decimal numbers enclosed in parentheses, formatted like standard geographic coordinates.

Mr-Worldwide

Medium

Cryptography

picoCTF 2019

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Description

A musician left us a [message](#). What's it mean?

Hints ?

(None)

16,450 users solved

35% Liked

picoCTF{FLAG}

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Analysis

The hint lies in the name "Mr-Worldwide," the moniker for the rapper **Pitbull**. This suggests that the solution is global in nature. The coordinates provided are:

Plaintext:

```
message (1).txt
File Edit View H1 H2 H3 B I S G A
picoCTF{(35.028309, 135.753082)(46.469391, 30.740883)(39.758949, -84.191605)(41.015137, 28.979530)(24.466667, 54.366669)(3.140853, 101.693207)(9.005401, 38.763611)(-3.989038, -79.203560)(52.377956, 4.897070)(41.085651, -73.858467)(57.790001, -152.407227)(31.205753, 29.924526)}
```

By inputting these into a mapping tool (like Google Maps) or a reverse geocoding API, we find that each coordinate corresponds to a specific city around the world. The "cryptography" element is a **Substitution Cipher** where each coordinate represents the **first letter** of the city located at those coordinates.

Decryption Table

Coordinates	City	Initial
(57.790000, -152.407227)	Kodiak, USA	K
(46.482525, 30.723309)	Odessa, Ukraine	O
(25.204849, 55.270782)	Dubai, UAE	D
(41.008240, 28.978359)	Istanbul, Turkey	I
(36.753769, 3.058756)	Algiers, Algeria	A
(24.860735, 67.001137)	Karachi, Pakistan	K
–	Underscore	–
(36.202103, 37.134262)	Aleppo, Syria	A
(6.524379, 3.379206)	Lagos, Nigeria	L

(31.945391, 35.928371)	Amman, Jordan	A
(-0.546686, 166.921041)	Yaren, Nauru	Y
(15.500654, 32.559899)	Khartoum, Sudan	K
(-23.698042, 133.880747)	Alice Springs, Australia	A

This looks like a typical picoCTF geolocation challenge where each coordinate corresponds to a city. Converting them:

First group:

- (35.028309, 135.753082) → Kyoto
- (46.469391, 30.740883) → Odessa
- (39.758949, -84.191605) → Dayton
- (41.015137, 28.979530) → Istanbul
- (24.466667, 54.366669) → Abu Dhabi
- (3.140853, 101.693207) → Kuala Lumpur

Second group:

- (9.005401, 38.763611) → Addis Ababa
- (-3.989038, -79.203560) → Loja
- (52.377956, 4.897070) → Amsterdam
- (41.085651, -73.858467) → Yonkers
- (57.790001, -152.407227) → Kodiak
- (31.205753, 29.924526) → Alexandria

Now take the **first letters** of each city:

- First group → **K O D I A K**
- Second group → **A L A Y K A**

So the decoded flag is:

picoCTF{KODIAK_ALAYKA}

⬇
("ALAYKA" looks like a slight misspelling of *Alaska*, which is likely intentional or part of the challenge.)

Conclusion

By stringing the first letters together according to the original sequence, we uncover the hidden message.

Flag: picoCTF{KODIAK_ALAYKA}

Reflection

This challenge is a classic example of **OSINT (Open Source Intelligence)** blending with cryptography. While the technical "encryption" is a simple substitution cipher, the difficulty lies in identifying the **external data source** required to decode it.

- **Key Takeaway:** In CTFs, the challenge title and "flavor text" are rarely just decoration. The "Mr. Worldwide" hint was the essential pivot point that turned a wall of random numbers into a map-reading exercise.
- **Methodology:** Efficiency in this challenge depends on using the right tools. While looking up cities manually works, a script utilizing a reverse geocoding API (like Nominatim) would be the "pro" way to solve this quickly, especially if the coordinate list were longer.
- **Nuance:** Geographic coordinates can be imprecise. Depending on the map provider, one might get a suburb or a province name instead of a city. The "Aha!" A moment comes when you see the letters starting to form a recognizable word or pattern, allowing you to self-correct and verify the data.